

Section 1

Neuroanatomy

Sujin Park

COGS 17

10/03/25

Sujin Park

Education:

2nd year PhD Student in Cognitive Science
MA Psychology (Cognitive Neuroscience)
BA Political Science and Diplomacy & Psychology

Research:

- Biomarkers of Neurodevelopmental disorders

Contact Info:

- Email: sup031@ucsd.edu (Pls include COGS17 in subject line) or Canvas Inbox
- Discussion Section: Friday 9 & 10am (online)
- Office Hours: Mon 3:30-4:30pm (zoom link on the announcement on canvas)



Feel free to reach out if you have any questions or problems!

Ground Rules

Use this section to boost your learning

- The aim of this section is to review contents covered in class
- Attending section and actively engaging will improve your learning!
- It's okay to be wrong

Keep discussions on topic

- Everyone has different opinions about various things
- Let's keep the conversation about class and lecture subjects

Check these out:

- **Section slides are on [github](#) (scan QR code)**
- [Scribblesheet](#): 30s recap after the section where you can put your random thoughts and e-meet your classmates

Let's all get As!

Important Reminders

Homework Problem sets

- Homework problems are Required, they guide your learning and will inform us of how you are doing for the lectures
- Due every Friday at Midnight (**EXCEPT** Oct 15, Nov 26)
- IMPORTANT: NO LATE HOMEWORKS WILL BE ACCEPTED

Exams

- 4 Exams total: Online, Open book, “one shot” for consecutive 80 Minutes
- 3 Midterms are NON-cumulative
- 1 Final is Comprehensive (on the SAME DAY after 3rd Midterm)

Extra Credit (Max 20 points)

- SONA
- Mnemonics
- Homeworks

Anatomy and Function of the Brain

Nervous System
Cells

Vision, Audition, other sensory systems

Movement

Sleep, Sexual Behavior

Learning

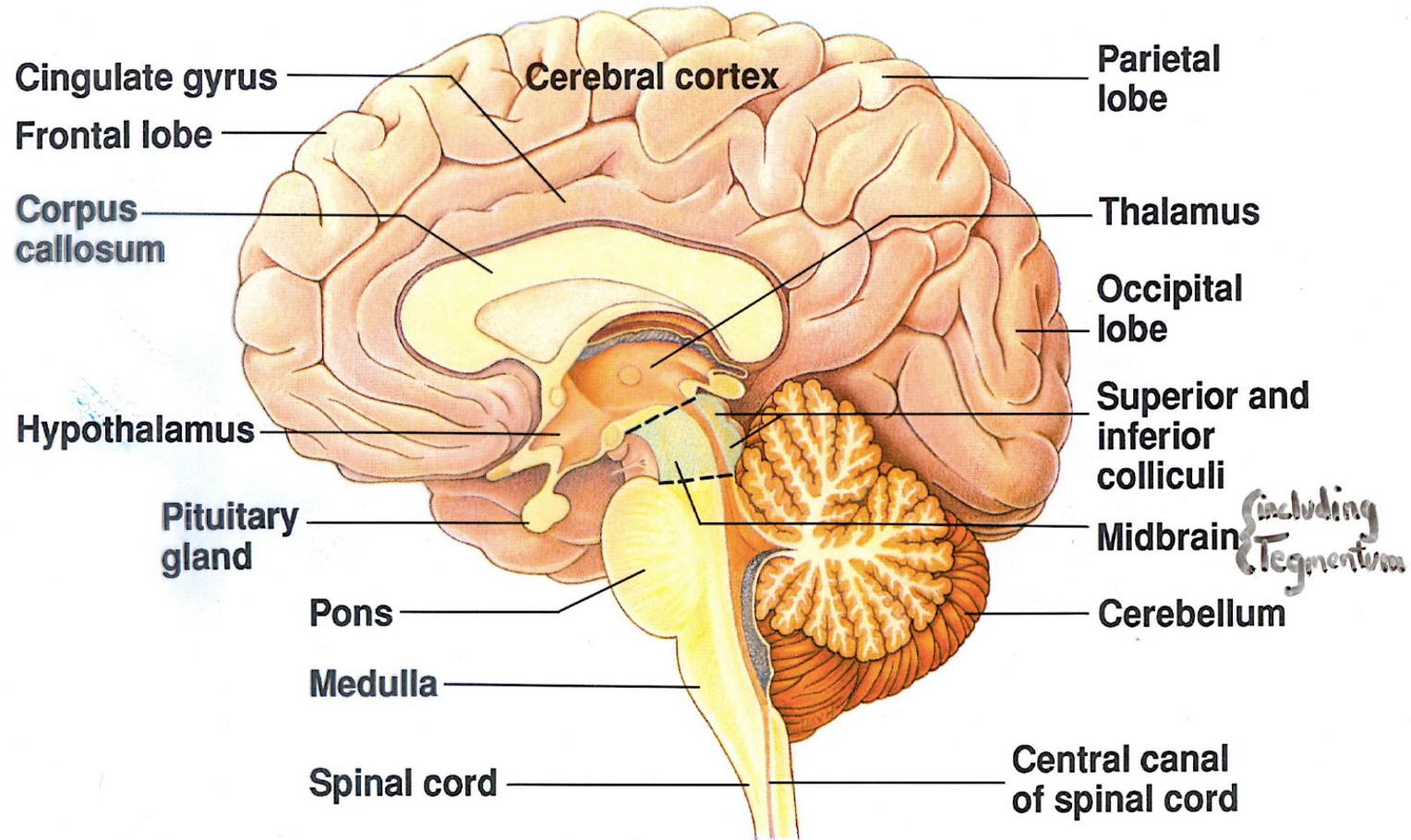
Language

...

Lecture 1

Anatomy of the Nervous System

Mid-Sagittal Section



Sagittal section of human brain

After Nieuwenhuys et al., 1988

© 1992 Wadsworth, Inc.

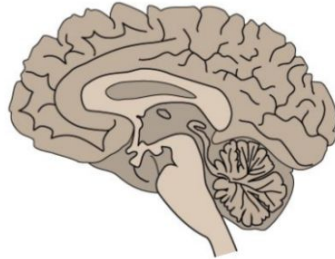
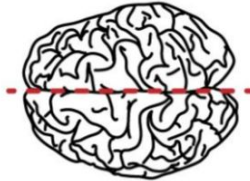
Planar Views of the Brain

Frontal or
coronal plane



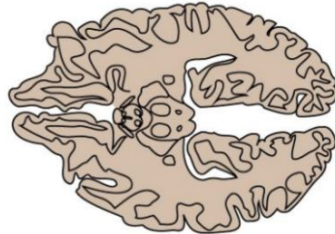
Coronal Plane -- From the **FRONT**

Sagittal plane



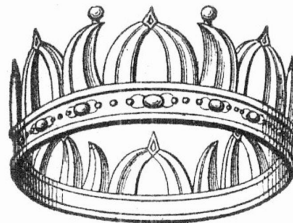
Sagittal Plane -- From the **SIDE**

Horizontal plane



Horizontal Plane -- From the **ABOVE**

Corona navalis (naval crown)



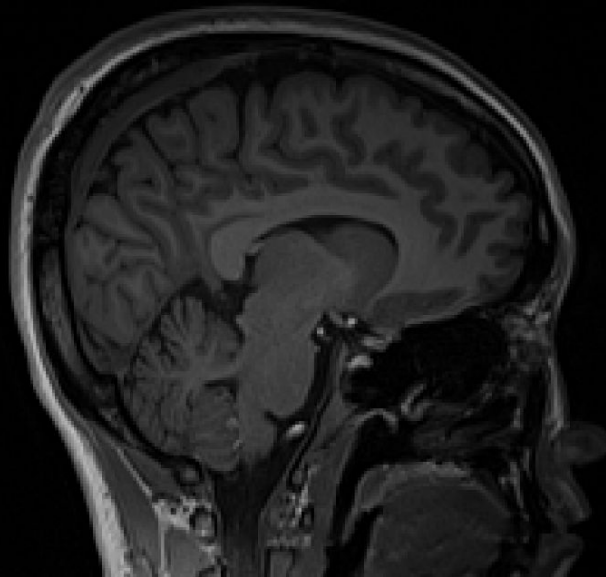
SAGITTARIUS



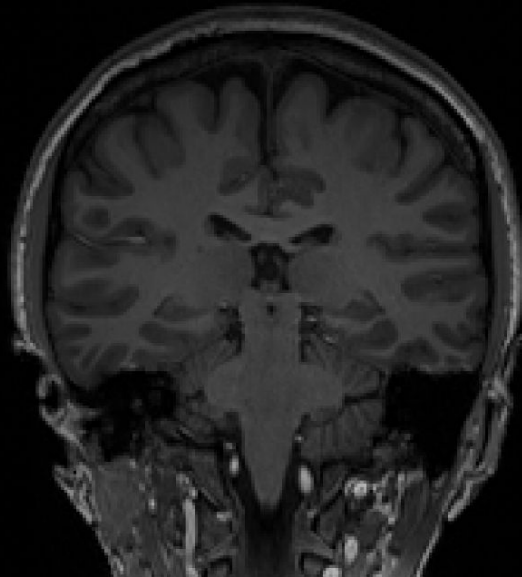
dreamstime.com

Planar Views of the Brain

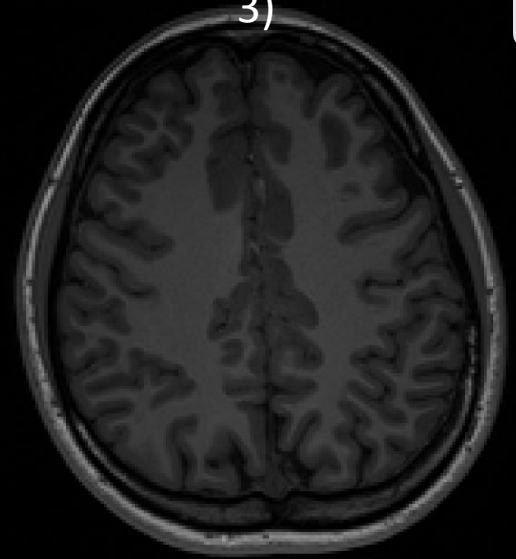
1)



2)



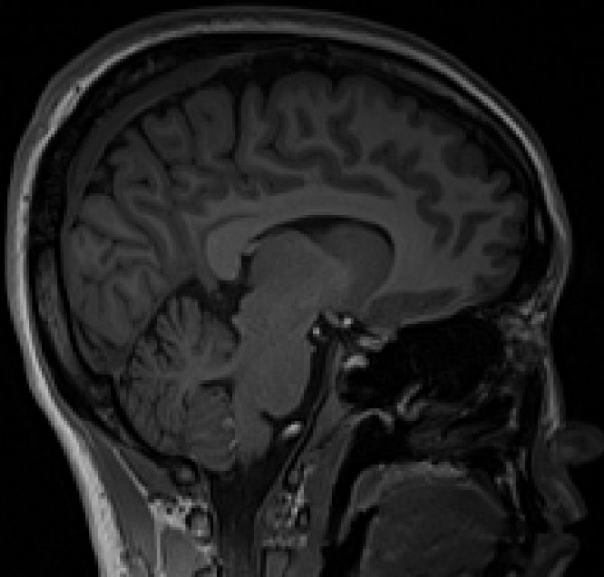
3)



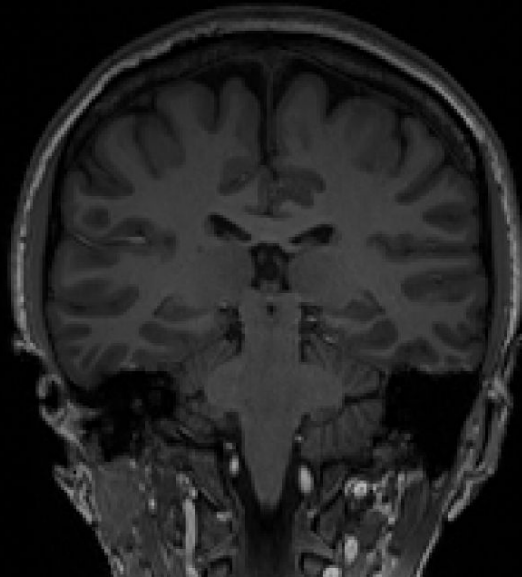
Guess whose brain this is 🤔

Planar Views of the Brain

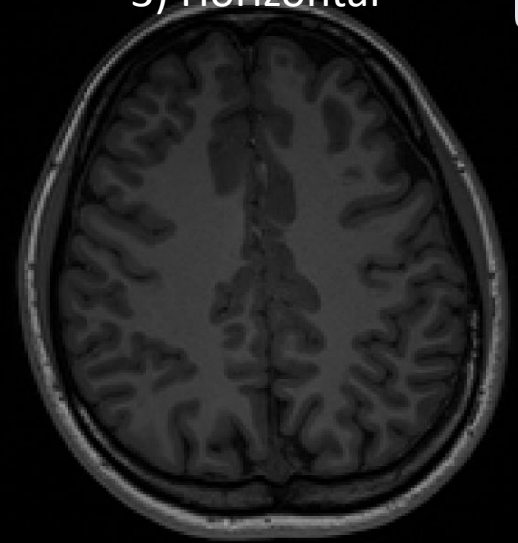
1) Sagittal



2) Coronal



3) Horizontal



Guess whose brain this is 🤔

Orientation and Views

Lateral & Medial

- Lateral: Towards the sides (Outside)
- Medial: Towards the middle (Center)

Dorsal & Ventral

- Dorsal: The “top” of the brain
 - Ventral: The “underside” of the brain
- * note dorsal = “rear”, ventral = “front” for the body

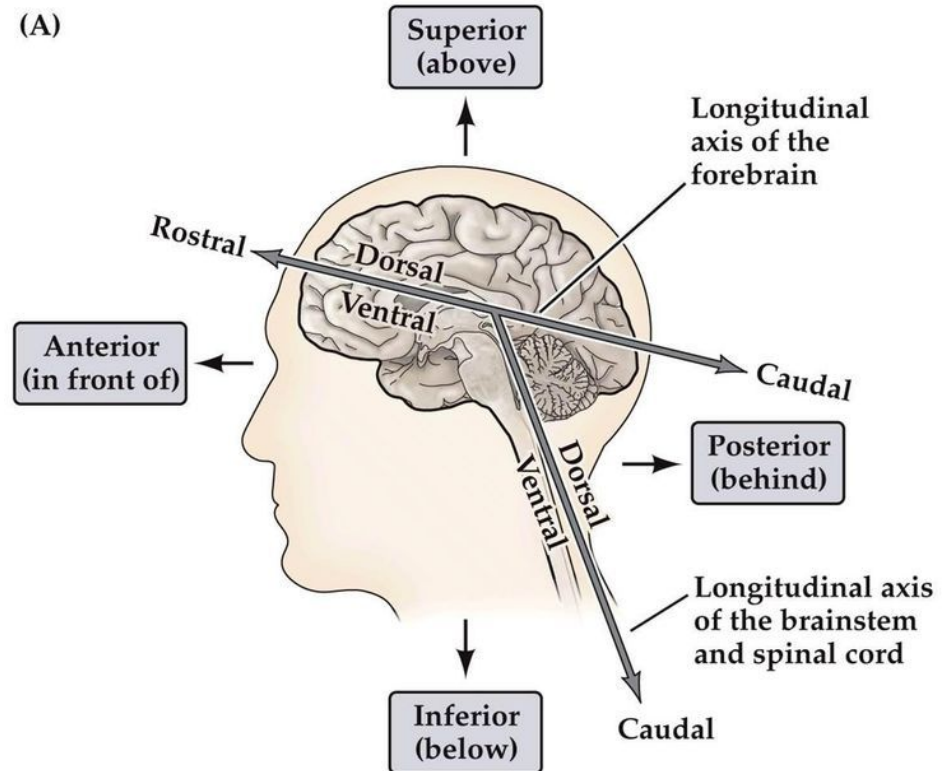
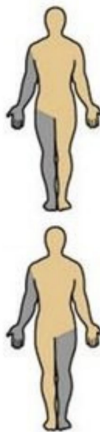
Anterior & Posterior

- Anterior: Front
- Posterior: Back

Bilateral Structure

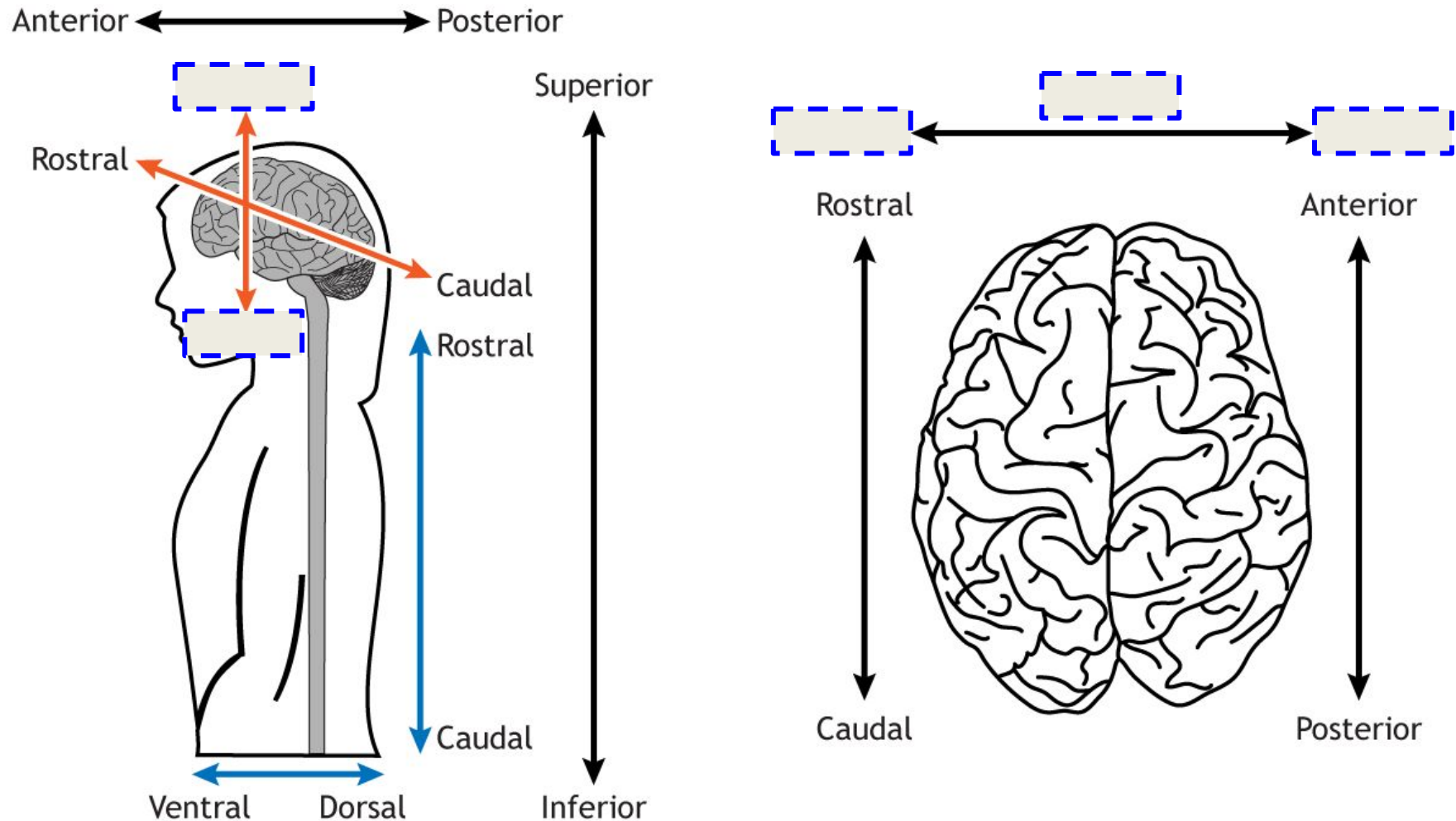
- Ipsilateral: Same side
- Contralateral: Opposite side

e.g., Left motor cortex controls right side of the body



NEUROSCIENCE 5e, Figure A1 (Part 1)
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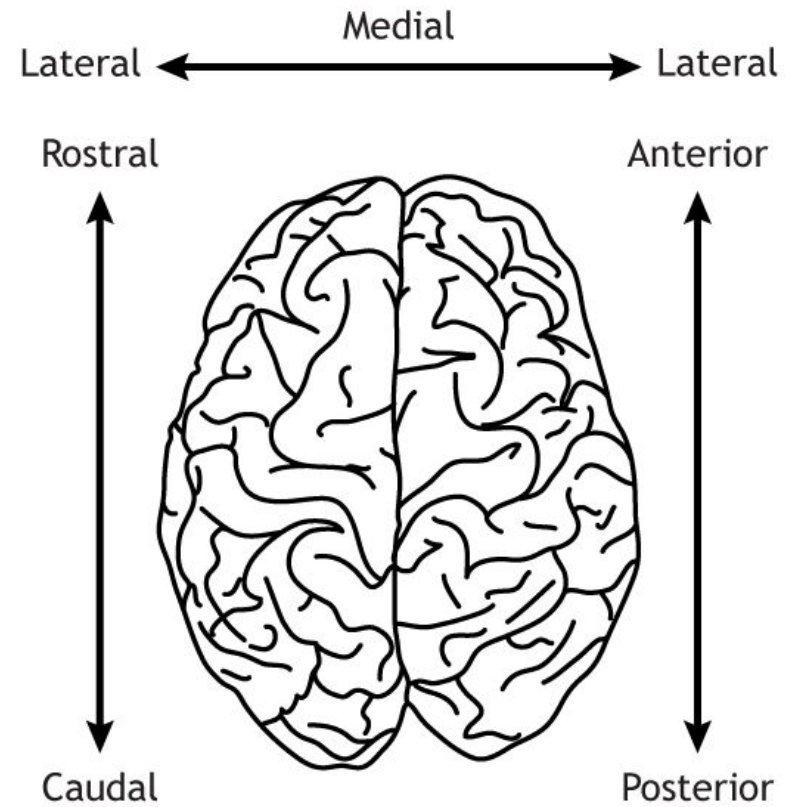
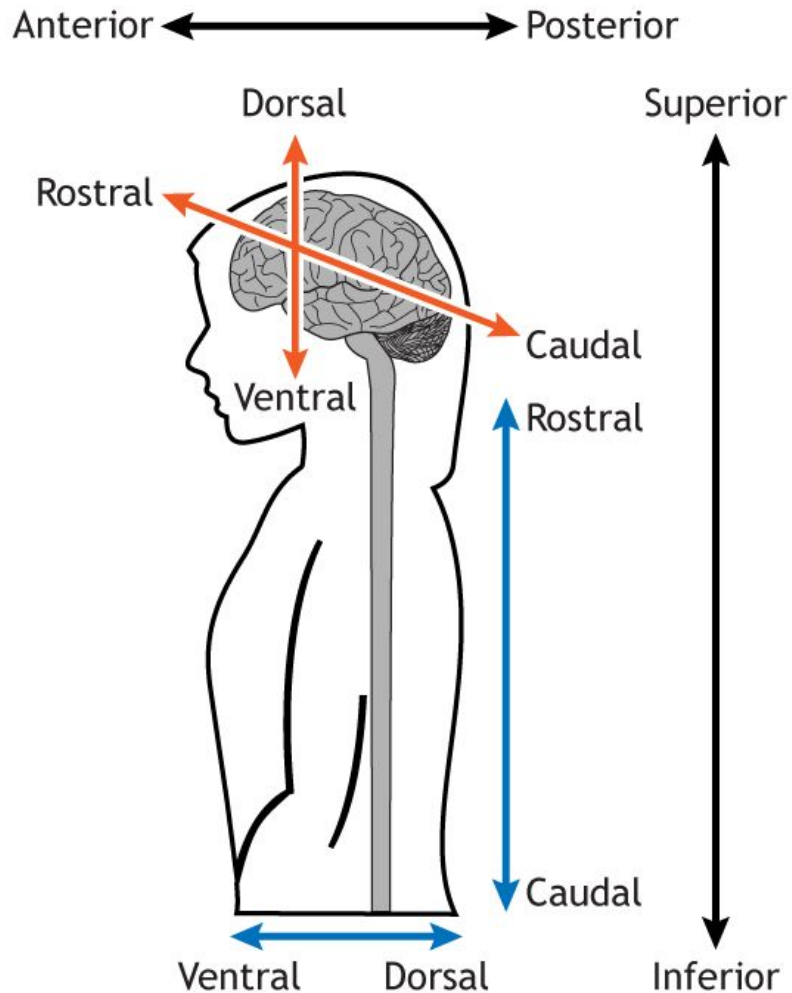
Orientation and Views



<https://openbooks.lib.msu.edu/neuroscience/chapter/anatomical-terminology/>

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Orientation and Views



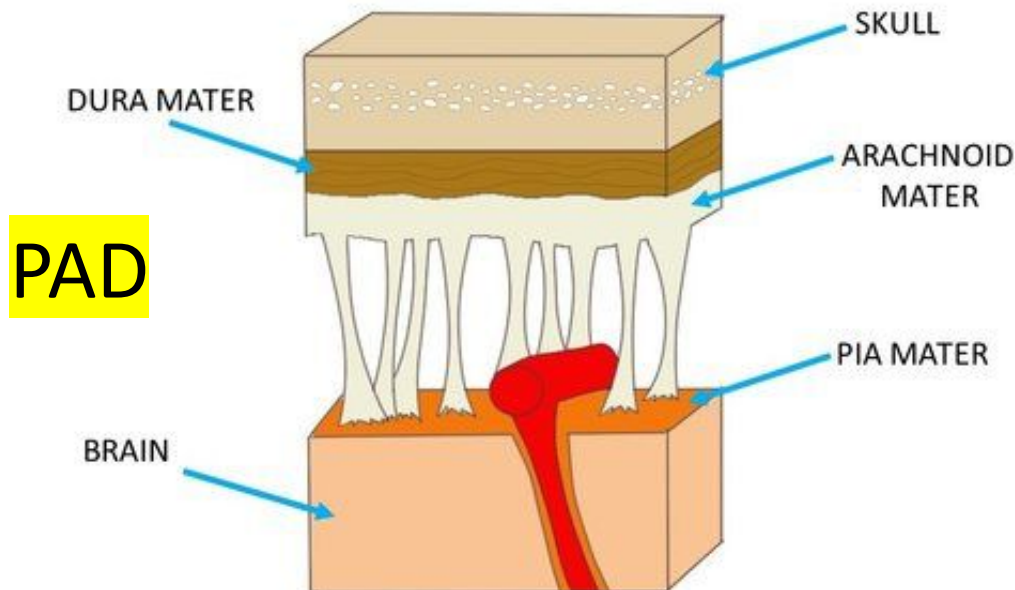
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Support Structure

Meninges: surrounds CNS under bone with 3 main layers

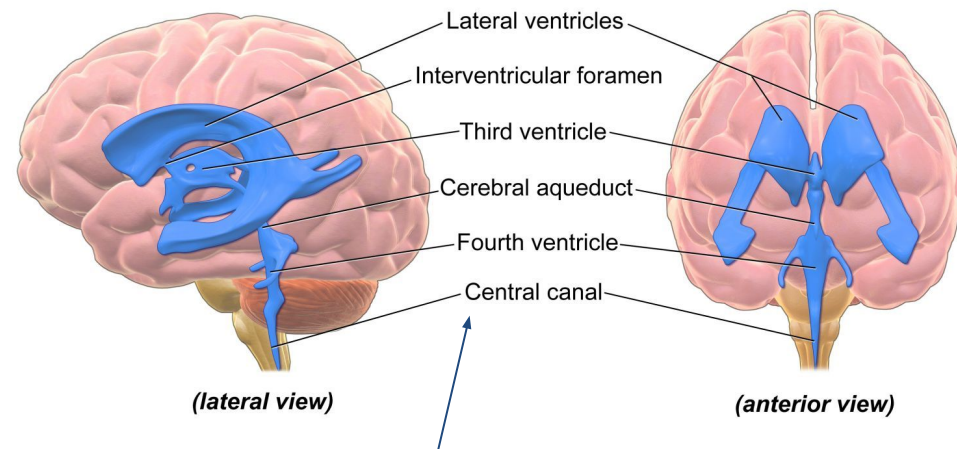
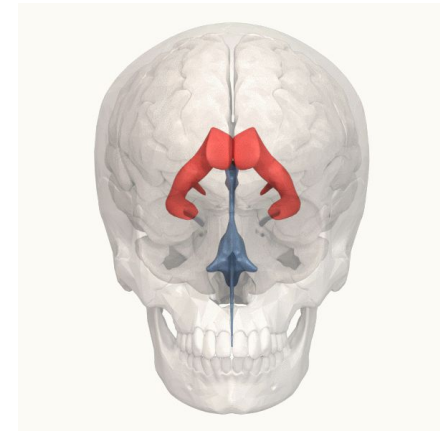
- Dura Mater (=Tough mother): Thick outer layer
- Arachnoid Mater: Spider-web like, spongy layer filled with Cerebrospinal fluid (CSF), shock absorber
- Pia Mater: flexible inner layer that conforms to the brain and spine surfaces, include blood vessels



Ventricles

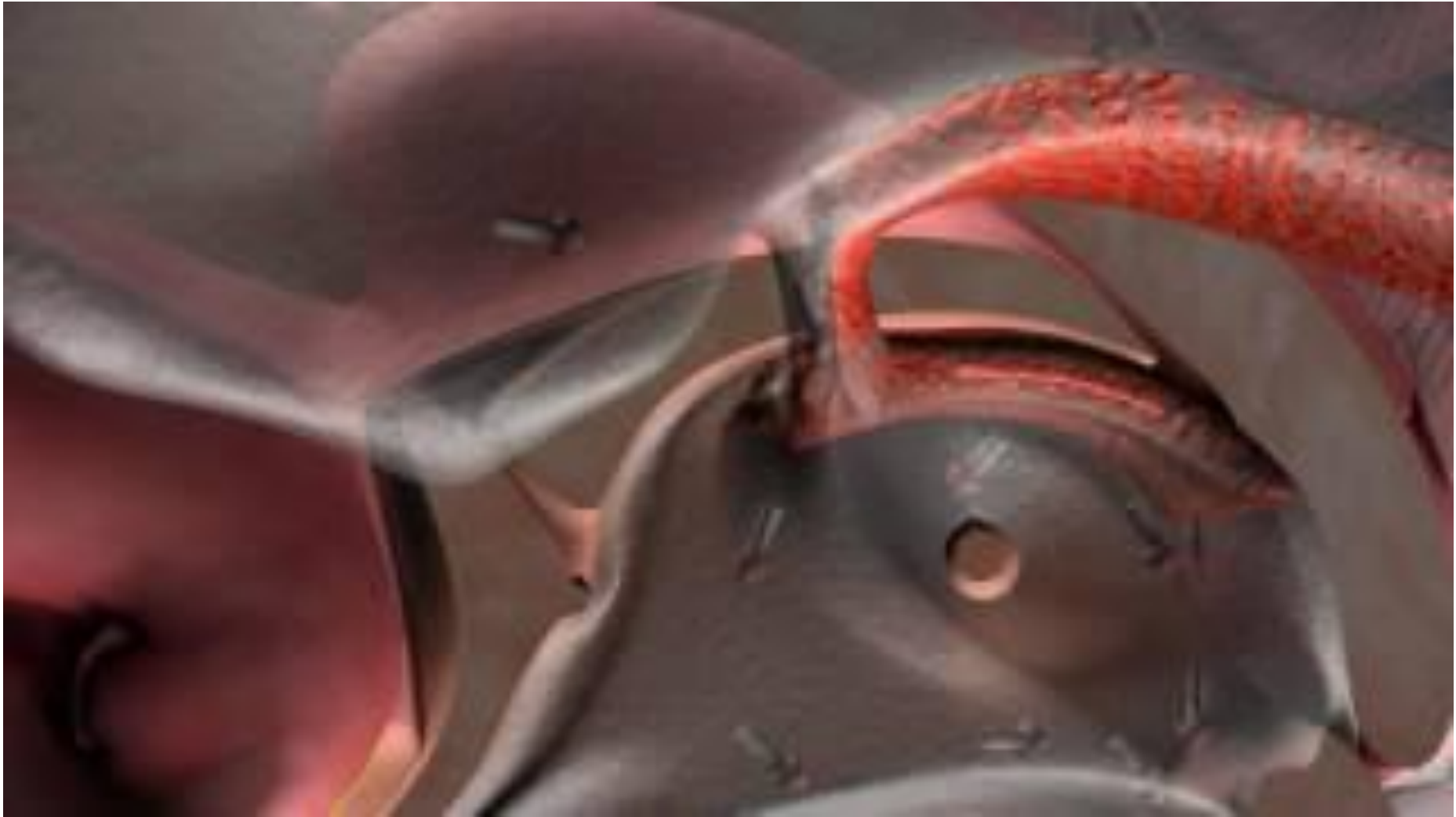
- Hollow (but not empty!), interconnected cavities
- produce and filled with CSF (also circulate it)
- Structure:

- 2 Lateral Ventricles
- Central Third Ventricle
- Cerebral Aqueduct
- Central Fourth Ventricle



only structure that is not duplicated on either side of the brain

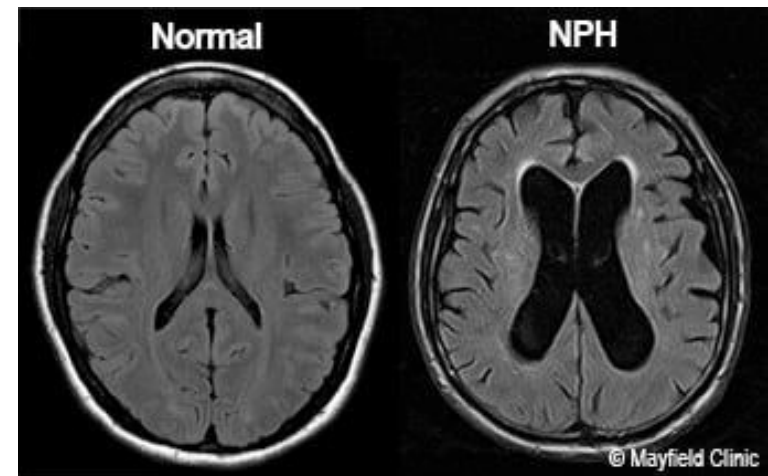
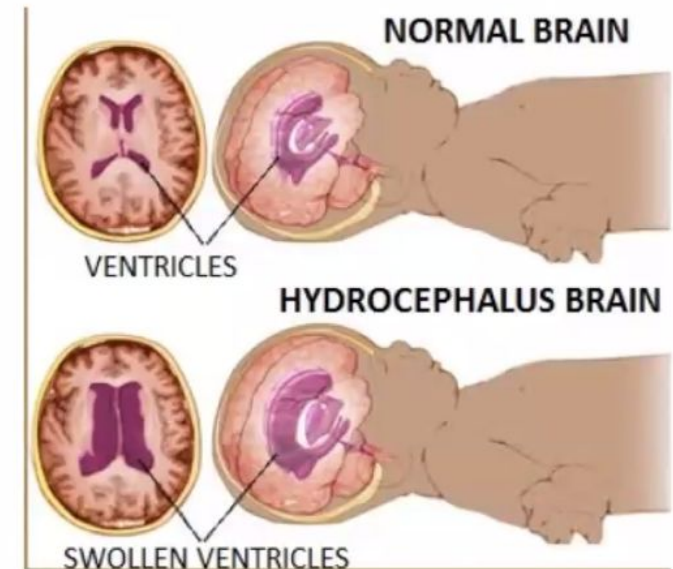
Support Structure



Hydrocephalus

Swollen ventricles

- When does this happen: CSF is not properly drained out through the cerebral aqueduct, the ventricles tend to swell up
- Then... CSF swelling pushes the brain matter against the PAD, replacing cortical matter with CSF
- Typically Fatal
 - Interventions can redirect excess CSF into the abdominal cavity to reduce the swelling





Feeding the brain

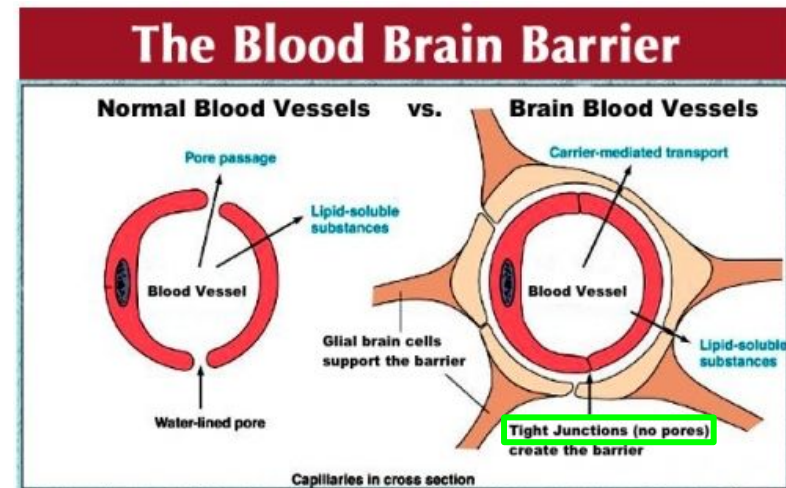
Blood Vessels

- Web of incoming **arteries** and outgoing **veins** (together, capillaries)
- Helps clear out the brain of waste
- Carries out “used” CSF
- 2% of body weight but uses 20% of blood supply



Blood-Brain Barrier (BBB)

- Strict control over what enters brain from bloodstream (x free flow!)
- Protects the brain from infections
- Only small uncharged particles (O₂, CO₂) and some fat-soluble molecules can passively cross BBB
- Astrocytes (Type of Glia cells) works as further barrier and actively fetch whatever needed and put them in the blood vessel (e.g., sugar)



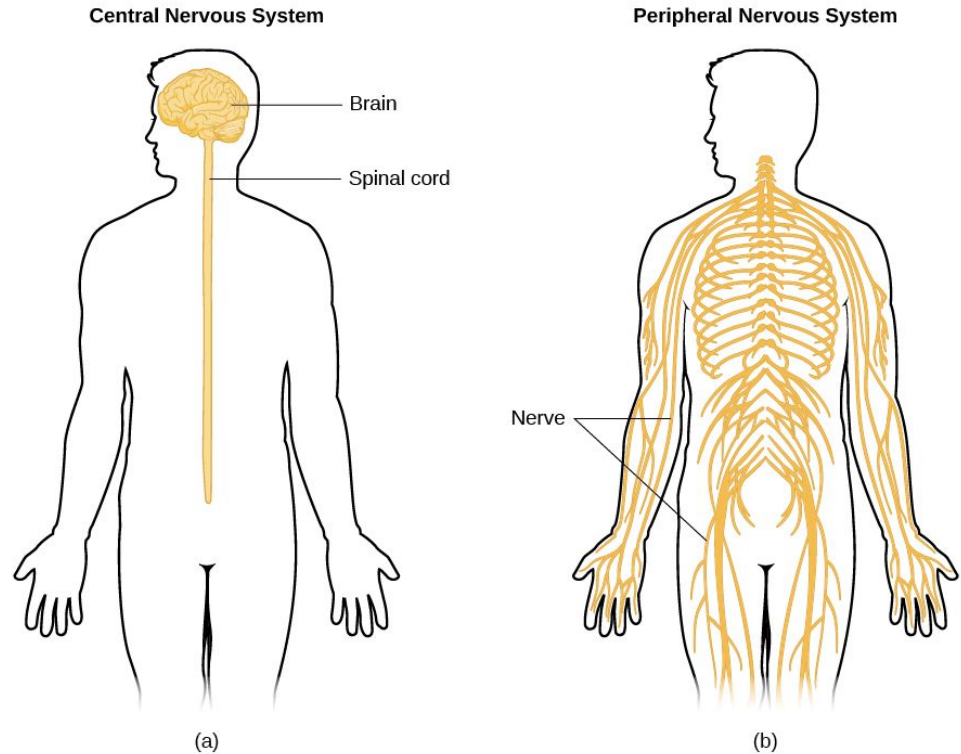
CNS vs PNS

Central Nervous System (CNS)

- Spinal cord and brain
- Encased in bone and meninges

Peripheral Nervous System (PNS)

- Nerves outside the CNS
- Somatic Nervous System:
interaction with the external
environment (Sensory/Motor)
- Autonomic Nervous System:
interaction with the internal
environment (internal organs)



Review Questions

Match definition to concept:

- | | |
|--|------------------|
| _____ Towards the center of the brain | A. Lateral |
| _____ Bottom of the brain | B. Medial |
| _____ Top of the brain | C. Dorsal |
| _____ On both sides of the brain | D. Ventral |
| _____ Same-side connections | E. Bilateral |
| _____ Towards the outside of the brain | F. Ipsilateral |
| _____ Opposite side connections | G. Contralateral |

Label as either pertaining to the central nervous system (CNS) or peripheral nervous system (PNS):

- | | |
|------------------------------|---------------------------------------|
| _____ Somatic nervous system | _____ Surrounded by bone and meninges |
| _____ Brain and spinal cord | _____ Autonomic nervous system |

Review Questions

Match definition to concept:

<u>B</u>	Towards the center of the brain	A. Lateral
<u>D</u>	Bottom of the brain	B. Medial
<u>C</u>	Top of the brain	C. Dorsal
<u>E</u>	On both sides of the brain	D. Ventral
<u>F</u>	Same-side connections	E. Bilateral
<u>A</u>	Towards the outside of the brain	F. Ipsilateral
<u>G</u>	Opposite side connections	G. Contralateral

Label as either pertaining to the central nervous system (CNS) or peripheral nervous system (PNS):

<u>PNS</u>	Somatic nervous system	<u>CNS</u>	Surrounded by bone and meninges
<u>CNS</u>	Brain and spinal cord	<u>PNS</u>	Autonomic nervous system

HindBrain

Medulla Oblongata

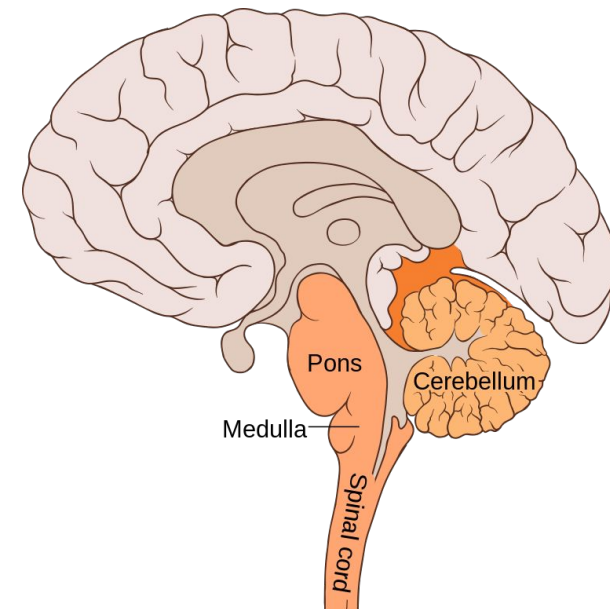
- Control primal reflexes (e.g., breathing, heart rate, coughing, vomiting)
- functions that basically keep us alive

Pons

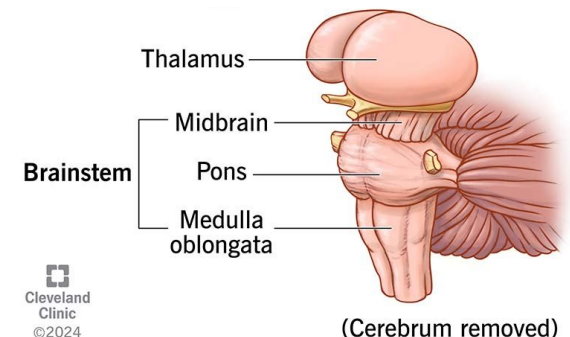
- Latin for “Bridge”: relay b/w cortex & cerebellum and brain & spinal cord
- Lateral: signals to/from cerebellum
- Medial: De-/Arousal system (i.e. reticular formation for arousal, and raphe system for sleep)

Cerebellum

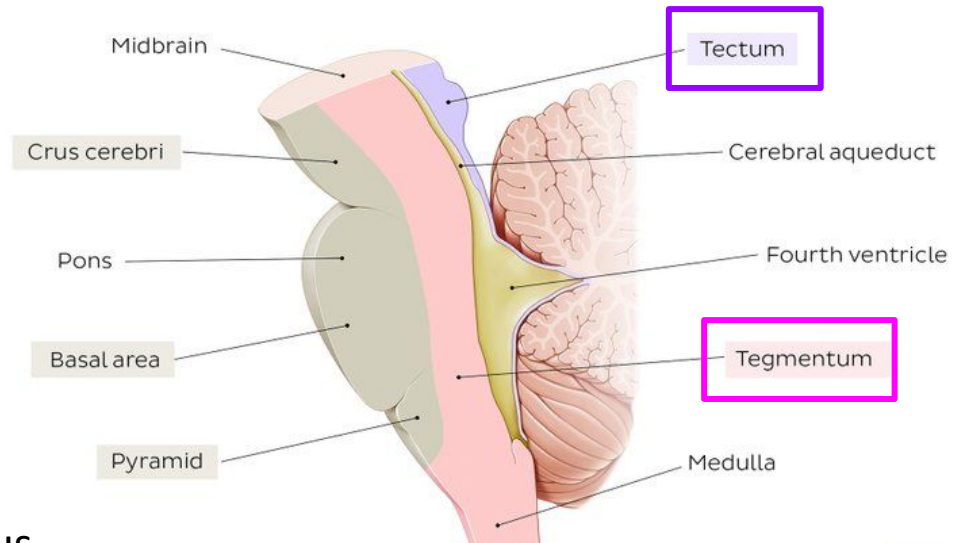
- Store motor programs w/ real-time sensory coordination
- Critical in timing/well-coordinated actions and also important for shifting attention
- Propose motor sequences and guide movements
- NOT the brain stem



+ Hypothalamus



Midbrain

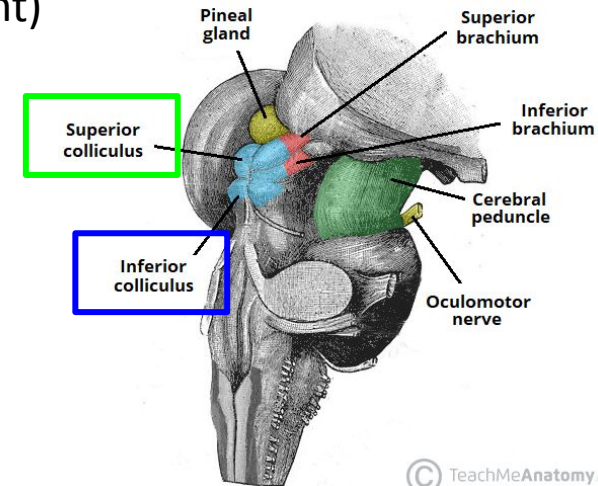


Tectum

- Latin for “Roof”
- Part of **sensory** pathways to the brain
- Consists of superior and inferior colliculus
 - 1) **Superior colliculus**: visual motion (including Blindsight)
 - 2) **Inferior colliculus**: auditory motion

Tegmentum

- Latin for “Covering” or “Rug”, below Tectum
- Contains major motor pathways and some cranial nerves
- Includes Red Nucleus and Substantia Nigra
- Contains cranial nerves to control eye movements



tectum to detect 'em, tegmentum for momentum

Forebrain (Diencephalon)

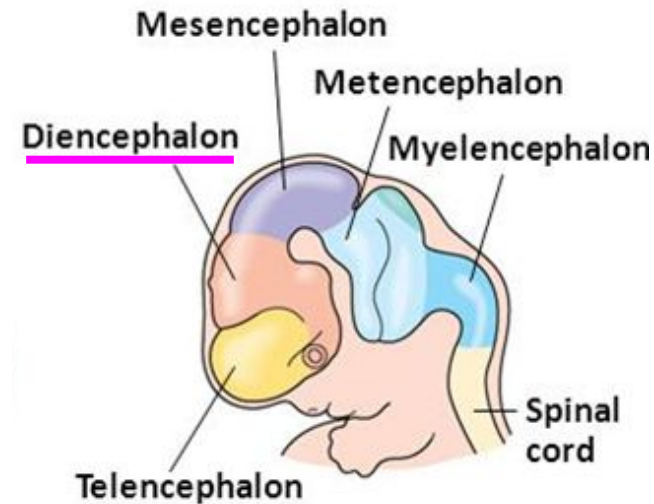
Diencephalon consists of the thalamus and hypothalamus

Thalamus

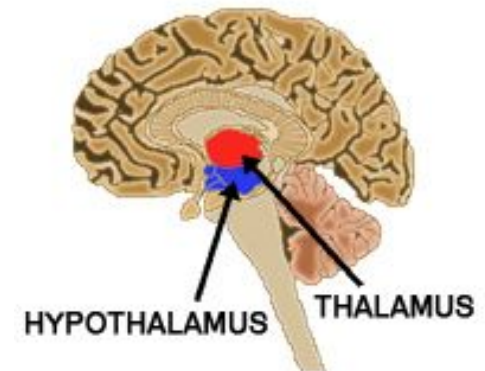
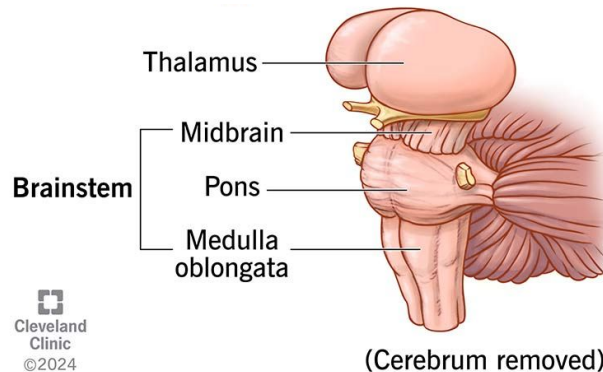
- Primary source of input to cerebral cortex
- Projects to/Receives from Sensory, Motor & Arousal sys
- Nuclei of many sensory and motor systems
- Involved in cortical arousal

Hypothalamus

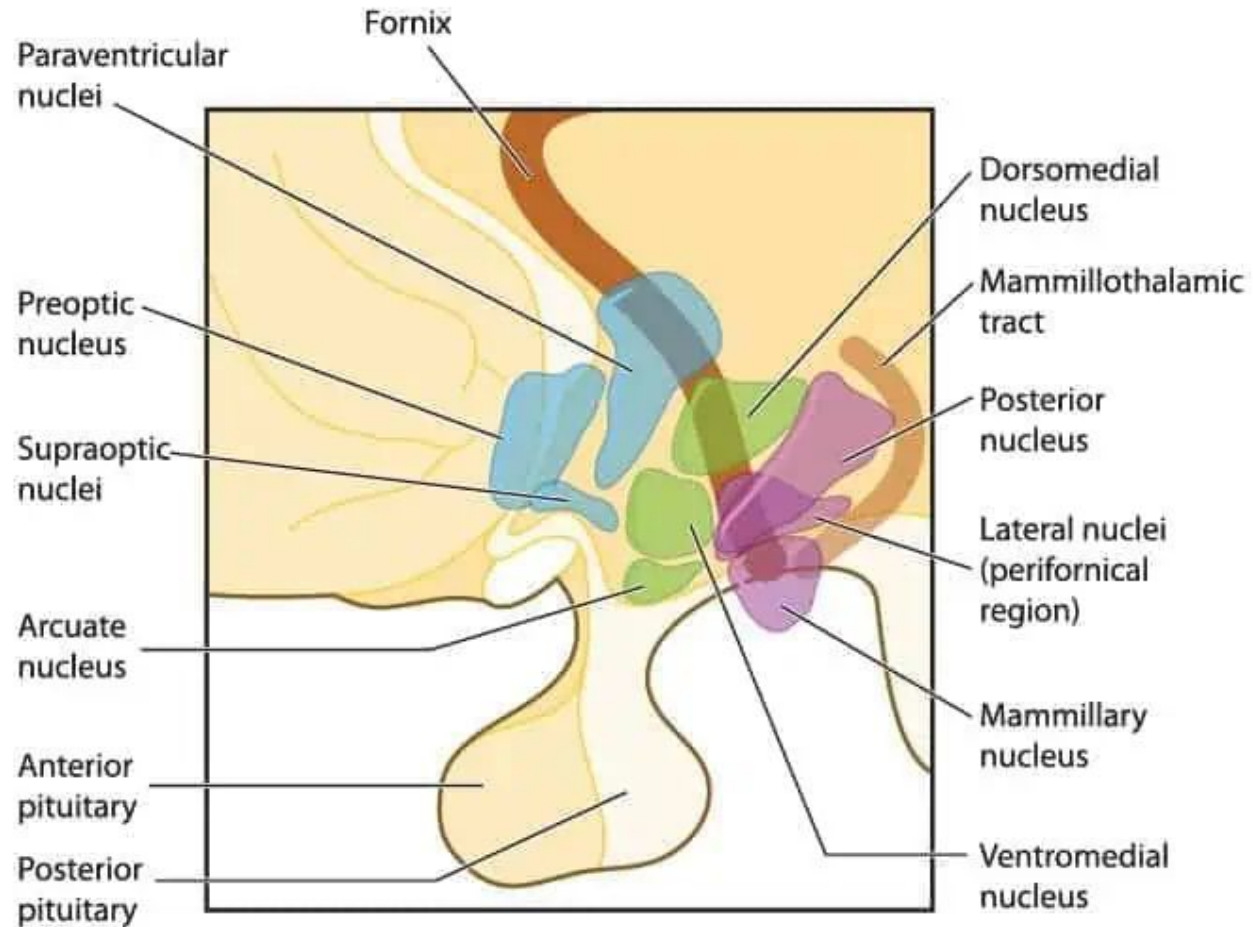
- Hypo = “below” → tucked in below Thalamus
- Oversees “4Fs”: Fighting, Fleeing, Feeding, F...
- Also regulates temperature and internal clock
- Neuro-Endocrine Sys: control brain + hormone sys by communicating to Pituitary Gland



Embryo at 5 weeks



Forebrain (Diencephalon)



nuclei of Hypothalamus

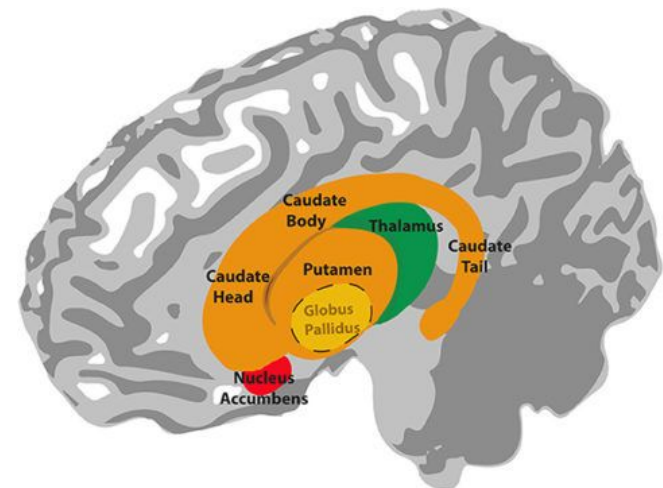
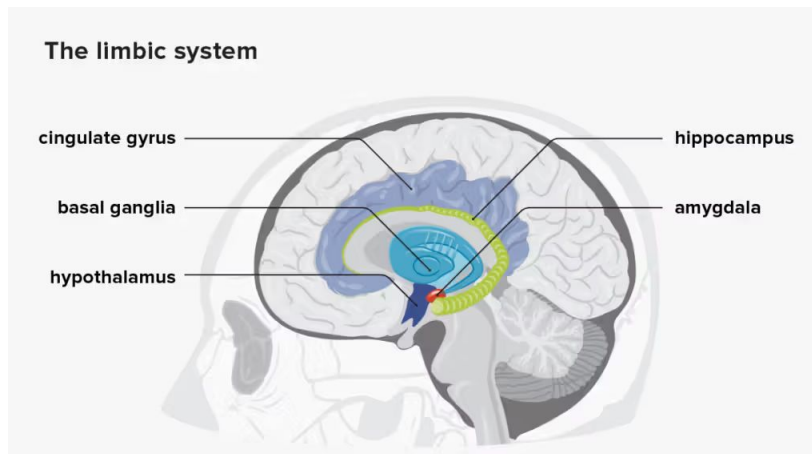
Forebrain (Telencephalon)

Limbic System

- Emotion, Motivation
- Hippocampus: formation of new memories and spatial mapping
- Amygdala: emotional expression and interpreting others' emotions
- Cingulate Gyrus: Valence +/- Evaluator, "RE-entrant System"
- Olfactory Bulb: exchanges olfactory information with the rest of the limbic system

Basal Ganglion

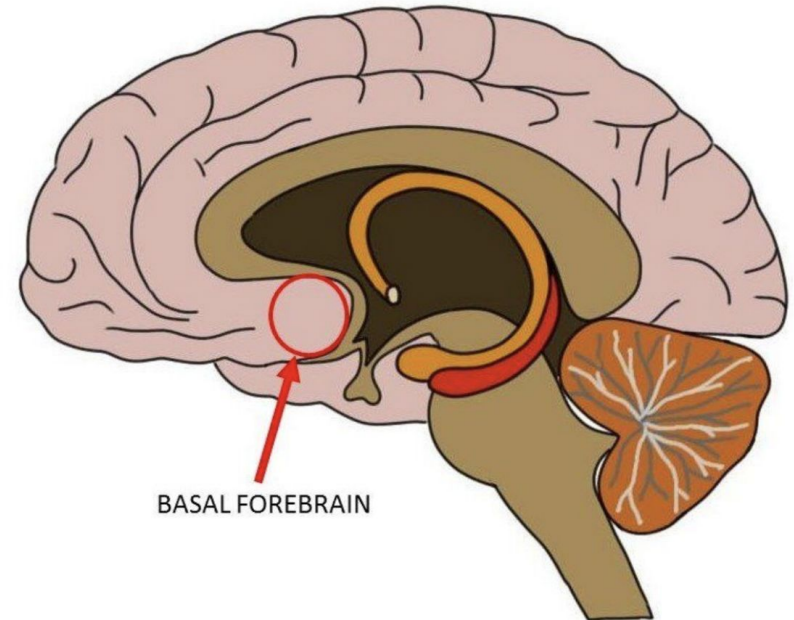
- Includes caudate, putamen, and globus pallidus
- "Motor Area", but diff. from cerebellum (which muscle to use): organizing activity into TASKS, especially **planned sequential behaviors** (task setting, check goals...)
- Another "RE-entrant system": bottom-up input + hierarchical cortical analysis



Forebrain (Telencephalon)

Basal Forebrain

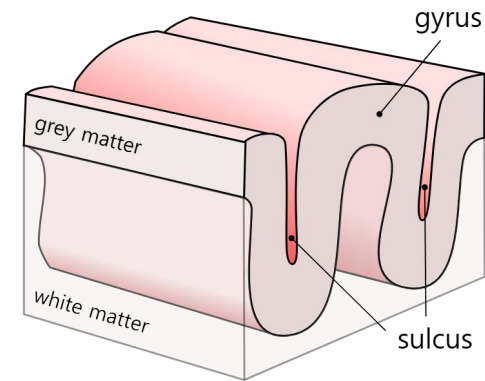
- Cortical area just anterior to Hypothalamus
- Major role in attention and cortical arousal
- Main source of ACh (Acetylcholine, excitatory neural transmitter; wakes you up in the morning...) and GABA (Gamma-Aminobutyric Acid, de-arousal/inhibitory neural transmitter; shut you down, put you to sleep...)
- Receives input from Raphe/Reticular Arousal System in Brainstem



Cerebral Cortex

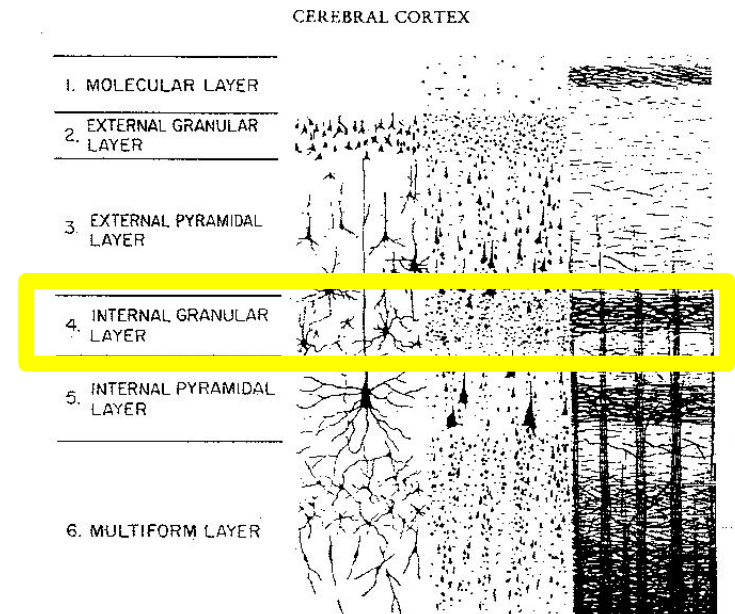
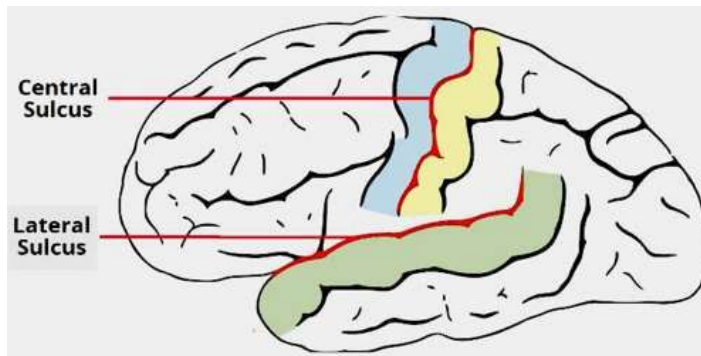
Organized into 6 layers (highly convoluted)

- Information projected to cortex enters at layer 4
- Bulges = gyri (sing. gyrus)
- Folds = sulci (sing. sulcus)

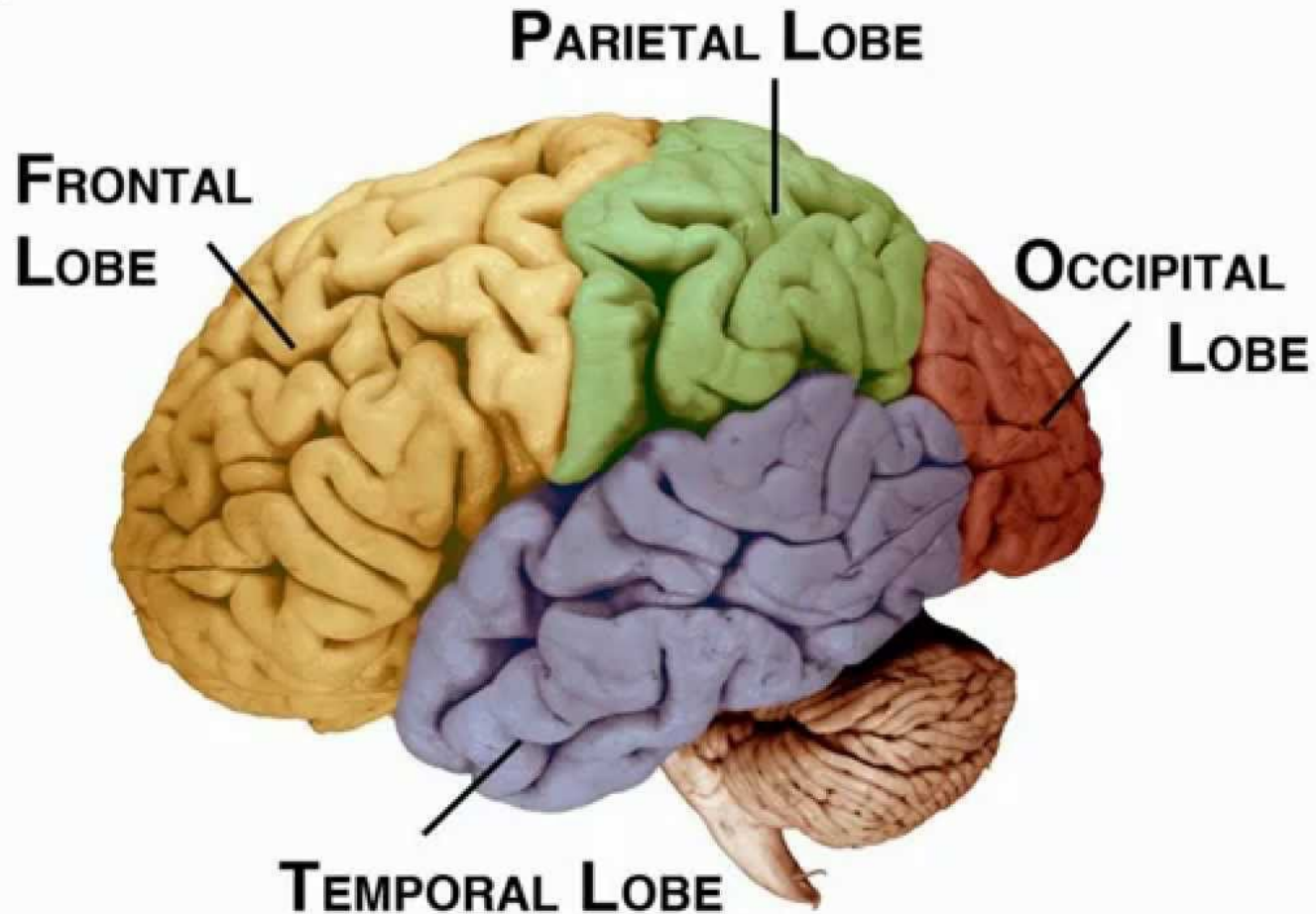


Landmarks

- Central Sulcus divides parietal from frontal lobe
- Lateral Sulcus/Fissure divides temporal from frontal lobe



Lobes of the Brain



Breaking down the Lobes

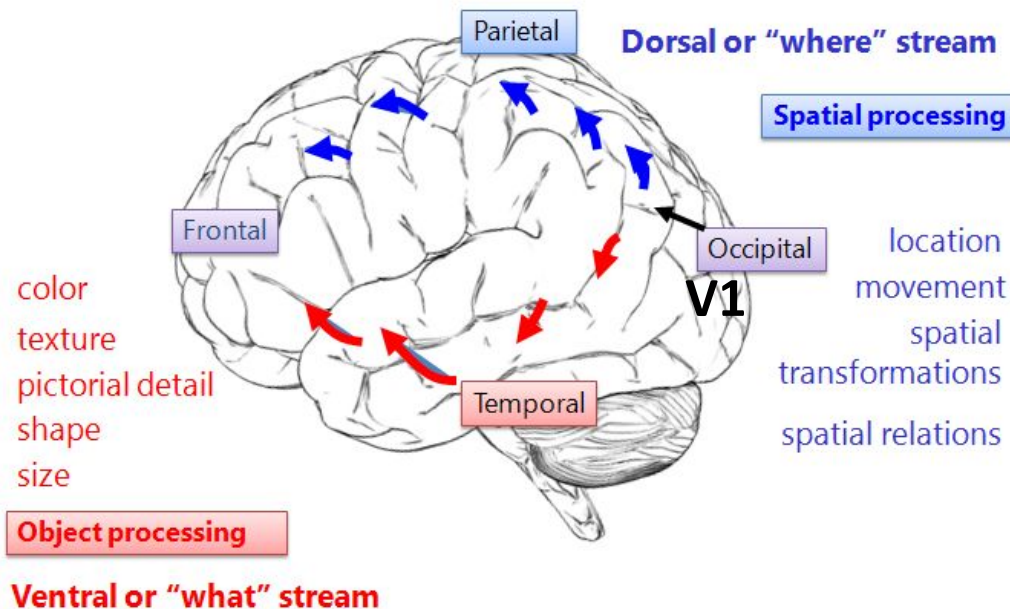
Occipital Lobe

- Devoted to Visual processing
- Contains V1 (primary visual cortex) and receives projections from thalamus
- major visual pathways: Ventral/Dorsal stream

Temporal Lobe

- Contains Medial Temporal (MT) - includes Direction-sensitive motion detectors, Medial Superior Temporal (MST) - includes Optic flow detectors, Ventral Visual Pathway - include Face cells,, Auditory areas - includes Wernicke's Area, and Anterior Temporal - emotional expression and interpretation (esp. right hem)

Ventral
= Who/What
: color & detail



Dorsal
Where/How
: Motion & Depth

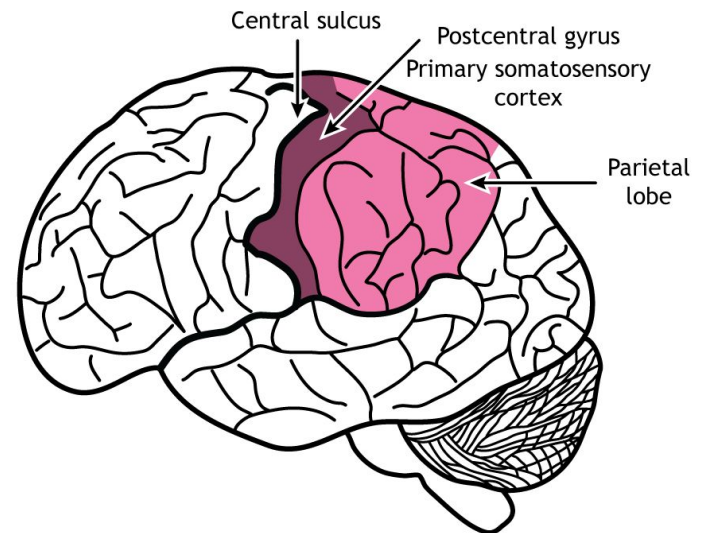
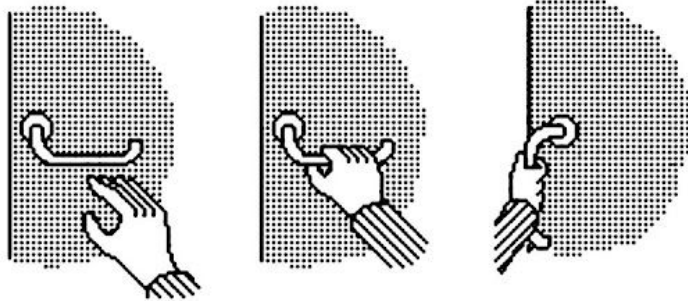
Breaking down the Lobes

Parietal Lobe

- integrating visual and somatosensory info (touch, pain, temperature, proprioception such as body position and movement, etc)
- Post-central gyrus (S1): primary somatosensory cortex
- also includes higher visual areas of “where/how” pathway, and the examples are...

Canonical cells

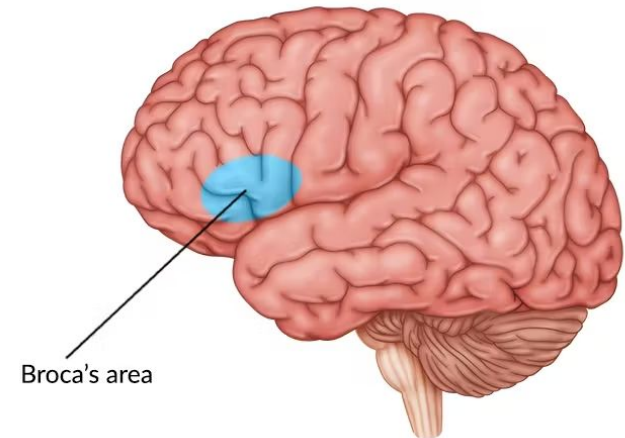
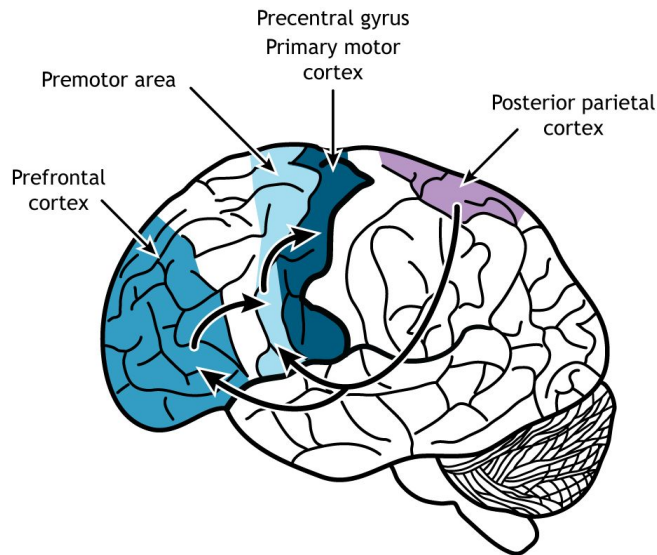
- Responds to “affordances” of objects (“Can I grab this object? How?”)
- Activity reverberates w/ premotor cortex, to shape how hand approaches



Breaking down the Lobes

Frontal Lobe

- Important for motor movements, language production, and strategy
- Precentral gyrus: primary motor cortex (initiating and controlling voluntary movements, “Ok, execute!”)
- Premotor areas: anterior to motor cortex (prepare to act → planning)
 - includes mirror cells (w/ Parietal) which responds to seeing self or other perform familiar manual tasks → simulation of observed action
 - includes Broca’s Area (activate just before speech production and articulation)
- Prefrontal cortex

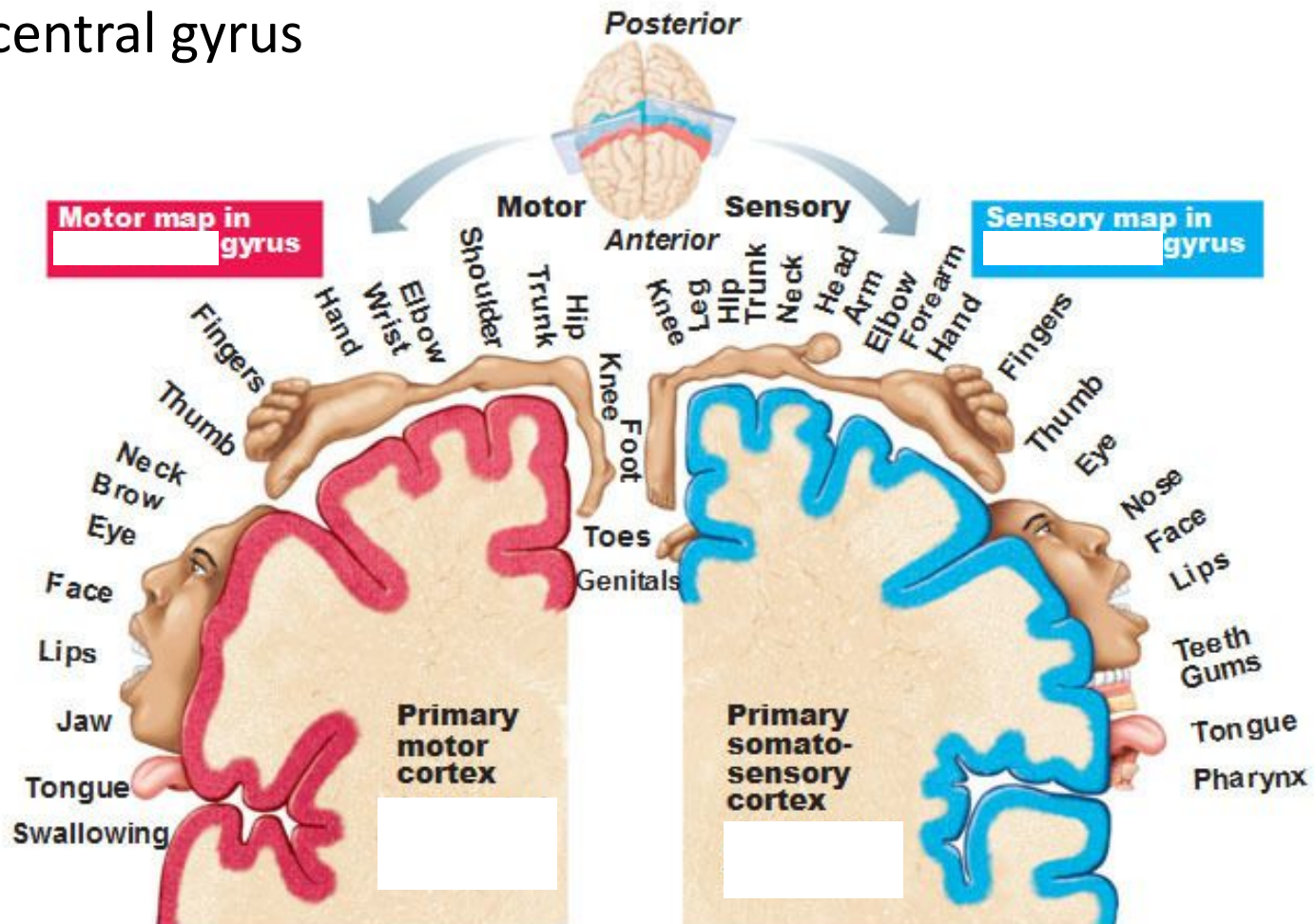


Breaking down the Lobes

Do you remember...?

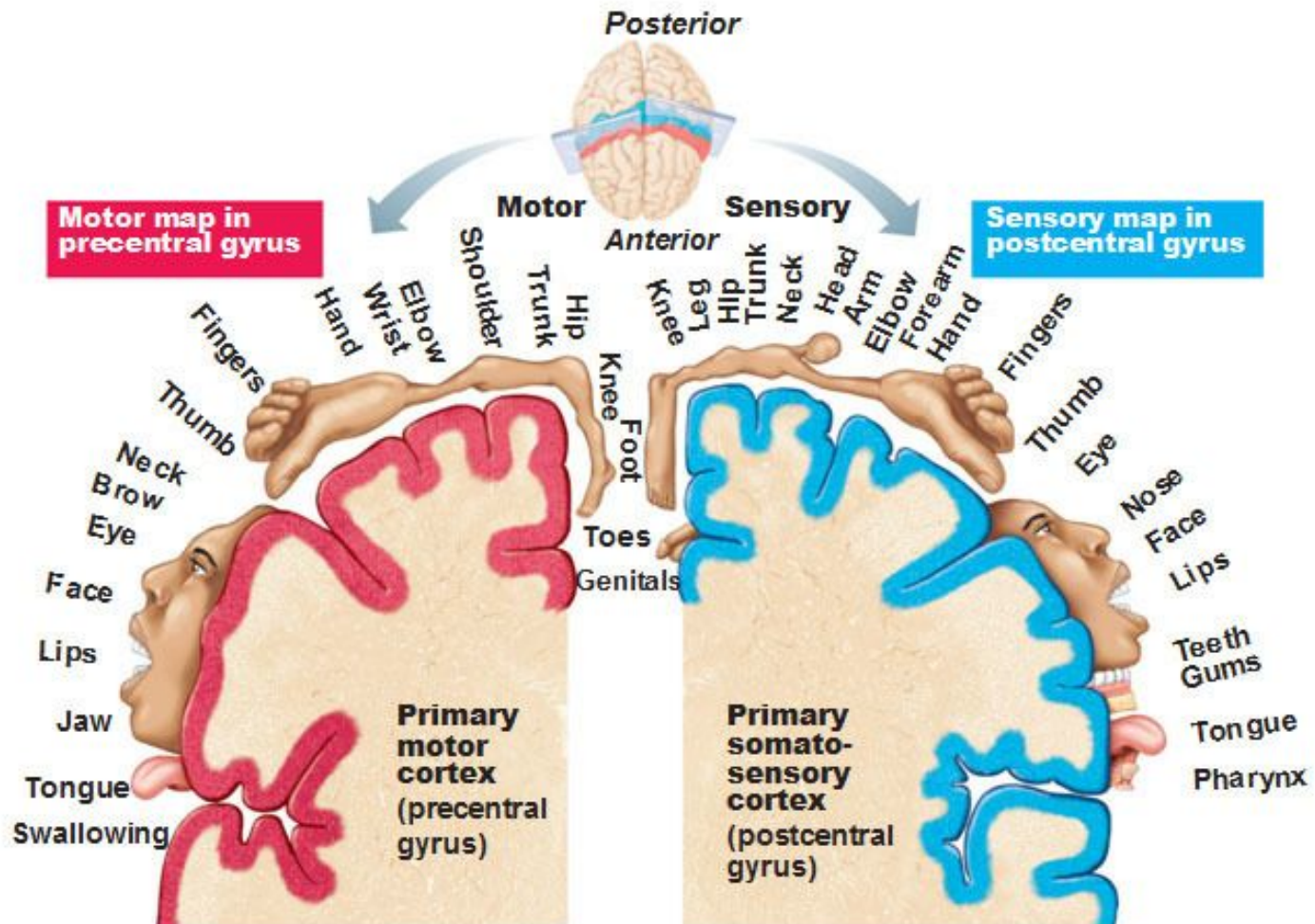
- precentral gyrus
- postcentral gyrus

Note that each hemisphere receives info from the opposite side of the body



Breaking down the Lobes

Note that each hemisphere receives info from the opposite side of the body

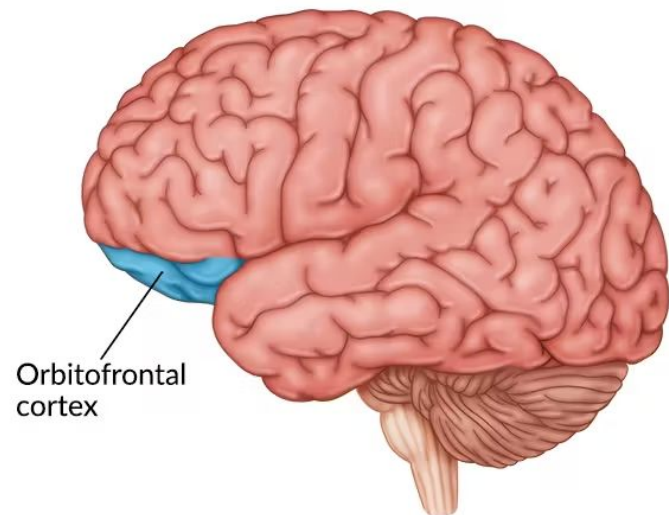
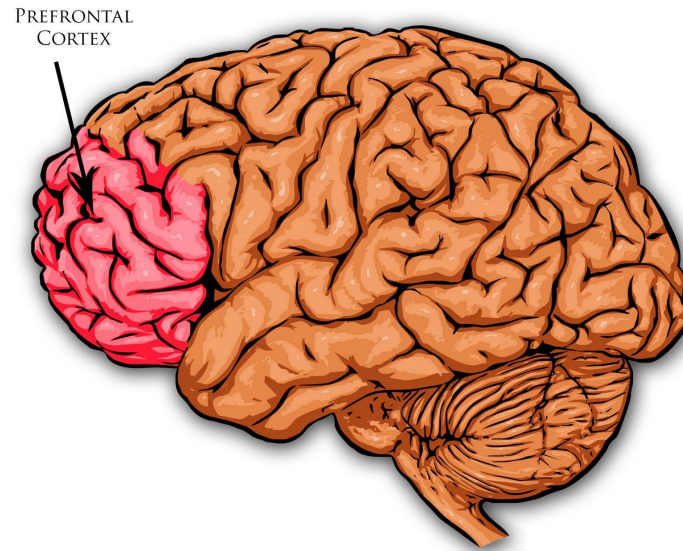


LEFT BRAIN VS. RIGHT BRAIN



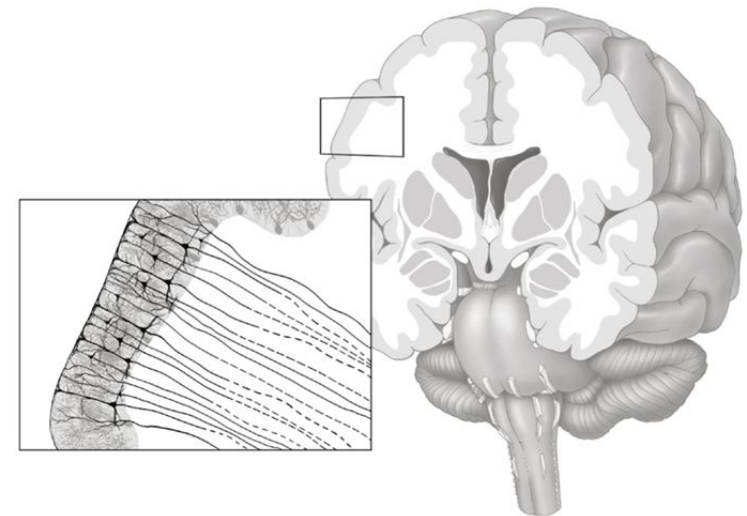
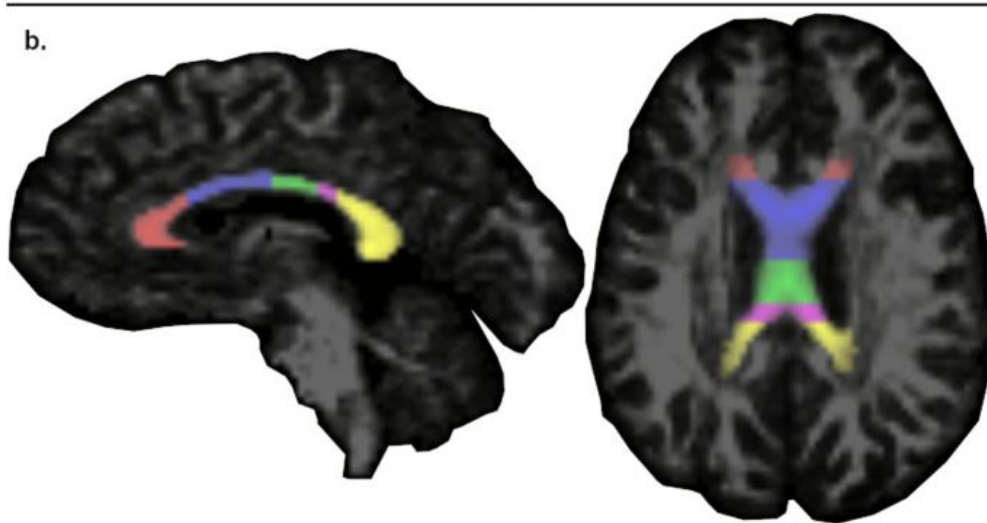
Prefrontal Cortex

- Self-control
- Delayed gratification
- Culturally/socially appropriate behavior
- Cost/benefit analysis
- Orbital-Frontal Cortex
 - Plays an important role in evaluating behavior of self and other, social strategy



Corpus Callosum

- Large axonal fibers connecting the two hemispheres
- Part of the “white matter” (connections b/w the little grey cells)
 - Consists of mainly myelinated axons
 - Brain = ~66% White Matter, by volume



<https://www.frontiersin.org/journals/psychiatry/articles/10.3389/fpsyt.2015.00035/full>

Spinal Cord

Consists of 31 segments along the spine

Dorsal (back part) Horns

- Afferent nerves
- Information from body to brain

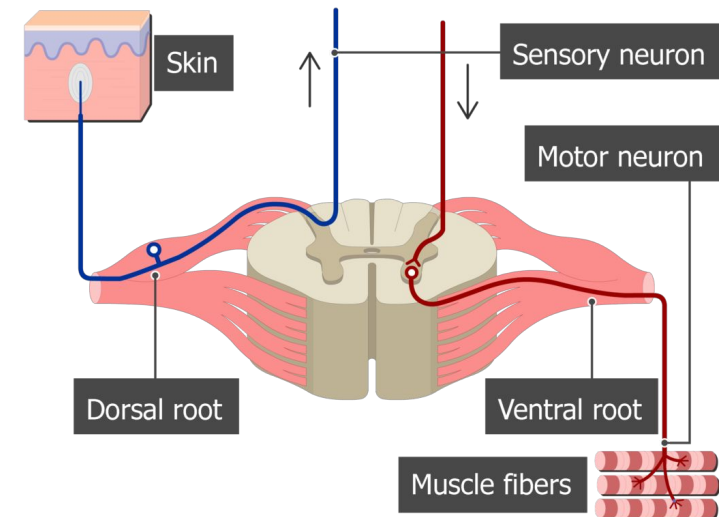
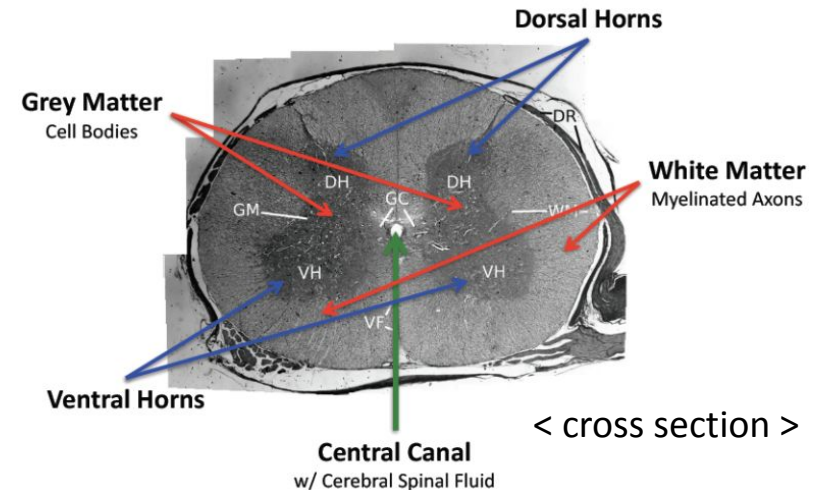
Ventral (front part) Horns

- Efferent nerves
- Motor information to muscles and glands

Bell-Magendie Law

- Sensory information enters (rear) dorsal horn
- motor information exits (front) ventral horn

→ *In the Door and out the Vent, that's how Spinal Cord info's sent*



Peripheral Nervous System

Somatic NS

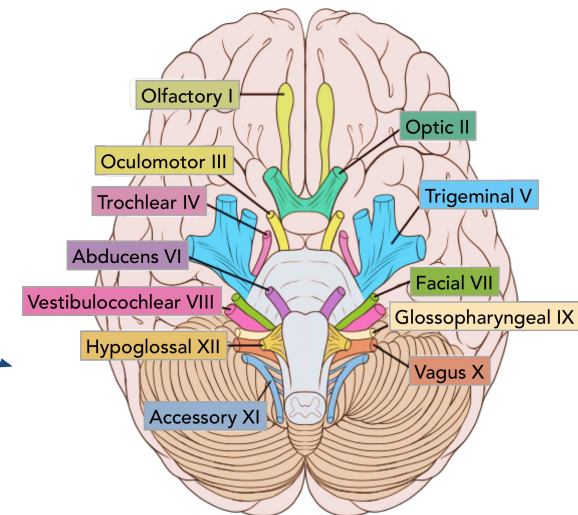
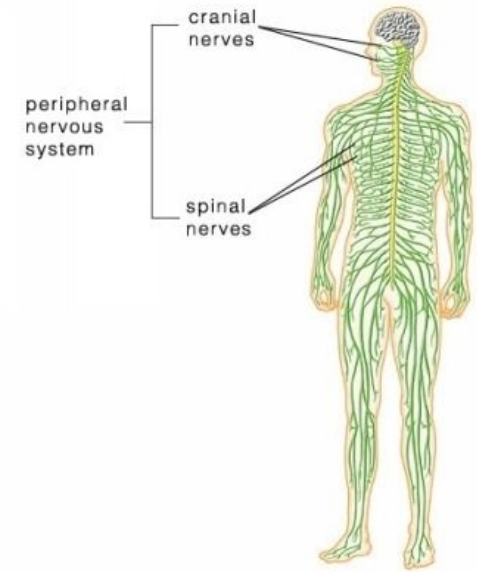
- 31 pairs of spinal nerves
 - Carry sensory input *from the* skin, muscles, and joints *to* the spinal cord.
 - Carry motor output *from* the spinal cord *to* the skeletal muscles.

Structure of each spinal nerve pair:

- Dorsal root (afferent): brings *sensory* signals into the spinal cord.
 - Ventral root (efferent): sends *motor* signals out to the muscles
-
- 12 cranial nerves: carrying sensory information from senses, receive feedback from some organs, control motor output to muscles that control eye movements, facial expressions

Autonomic NS

- Regulates internal state
 - 1) Sympathetic “Fight or Flight”
 - 2) Parasympathetic “Rest & Digest”



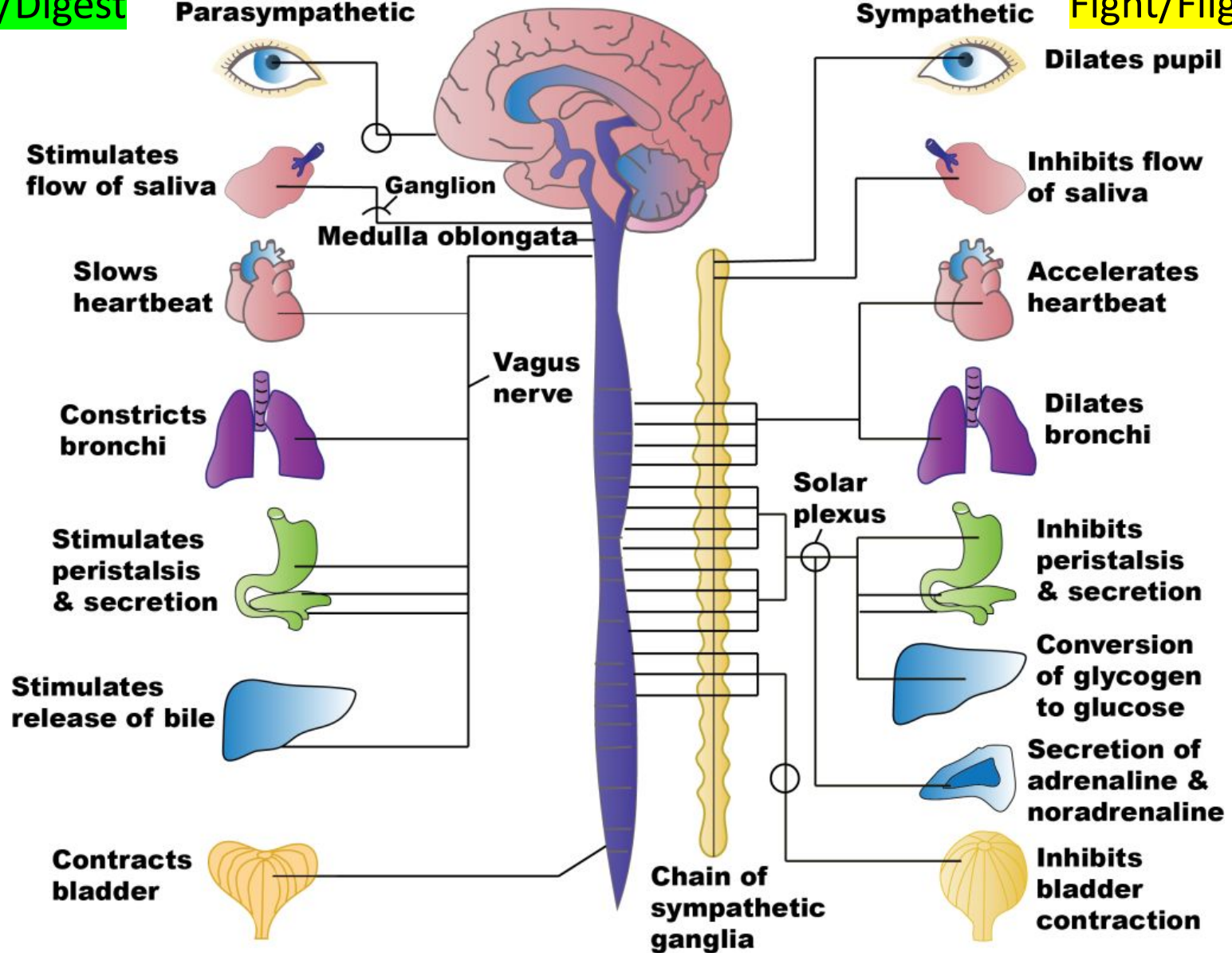
Autonomic Nervous System

Rest/Digest

Parasympathetic

Sympathetic

Fight/Flight



Review Questions

Label as either a sympathetic response (S) or a parasympathetic response (P):

- ___ Pupils dilate
- ___ Increased heartrate
- ___ Saliva production
- ___ Bronchi constrict
- ___ Halt digestion
- ___ Facilitate sexual arousal
- ___ Hold bladder
- ___ Blood vessels constrict
- ___ Pilo-erection

Review Questions

Label as either a sympathetic response (S) or a parasympathetic response (P):

- S Pupils dilate
- S Increased heartrate
- P Saliva production
- P Bronchi constrict
- S Halt digestion
- P Facilitate sexual arousal
- S Hold bladder
- S Blood vessels constrict
- S Pilo-erection

See you next week!

